Model Comparison Report

Project: Aerial Object Classification (Bird vs Drone)

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1. Objective

The objective of this project is to classify aerial images into **Bird** and **Drone** categories using deep learning models and compare their performance to identify the most effective approach.

2. Models Implemented

Custom CNN

- Built from scratch using convolutional layers
- Includes Batch Normalization and Dropout
- Learns features directly from dataset

ResNet50 (Transfer Learning)

- Pretrained on ImageNet
- Fine-tuned for binary classification
- Uses transfer learning for feature extraction

ResNet50 (Fine-Tuned)

- Pretrained on ImageNet
- Top layers unfrozen and fine-tuned
- Improved learning capability

3. Evaluation Metrics

The models were evaluated using:

- Accuracy
- Loss
- Precision, Recall, F1-score
- Overfitting Gap
- Training Time

4. Results Comparison

Metric	Custom CNN	ResNet50
Test Accuracy	**0.874**	0.828
Validation Accuracy	0.857	0.835
Loss	**0.344**	0.408
Precision (Bird)	**0.91**	0.84
Precision (Drone)	**0.84**	0.81
Recall (Bird)	0.87	**0.85**
Recall (Drone)	**0.88**	0.80
F1-score (Bird)	**0.89**	0.85
F1-score (Drone)	**0.86**	0.80
Overfitting Gap	0.044	**-0.041**
Training Time	**3.56**	4.93

5. Analysis

Custom CNN Performance

- Achieved **higher accuracy (87.4%)**
- Lower loss → better learning on dataset
- Better precision and F1-score for both classes
- Slight overfitting observed (gap = 0.044)

Indicates:

- Model fits training data well
- Slight generalization issue but still strong

ResNet50 Performance

- Lower accuracy (82.8%) compared to CNN
- Higher loss indicates less optimal learning
- Lower precision and F1-score
- Negative overfitting gap → **underfitting**

Indicates:

- Model did not fully adapt to dataset
- Likely due to:

    - insufficient fine-tuning
    - frozen layers limiting learning

Fine-Tuned ResNet50

- Significant improvement after fine-tuning
- Lower loss → better optimization
- Better generalization (less overfitting)
- More stable performance

Before fine-tuning → underfitting

After fine-tuning → strong performance

7. Conclusion

In this project, the Custom CNN model outperformed the ResNet50 model across most evaluation metrics, including accuracy, loss, precision, and F1-score.

After fine-tuning, ResNet performance improved significantly and achieved comparable accuracy to CNN, while providing better generalization and lower loss, making it more suitable for real-world deployment.